

Literature Status: Subspaces of Banach Lattices Without Bibasic Sequences

arXiv:1907.07589 answered by arXiv:2210.00614

2026-06-29

Status. `literature_already_answered`. This packet records a later-paper answer, not a new proof from this run.

Original source. M. A. Taylor and V. G. Troitsky, *Bibasic sequences in Banach lattices*, arXiv:1907.07589. In Remark 4.4, after showing that every closed infinite-dimensional subspace of the closed span of a bibasic sequence contains a bibasic sequence, the authors ask whether every closed infinite-dimensional subspace of a Banach lattice contains a bibasic sequence.

Supporting answer. T. Oikhberg, M. A. Taylor, P. Tradacete, and V. G. Troitsky, *Free Banach lattices*, arXiv:2210.00614. Its introduction says that Section 7 constructs the first example of a subspace of a Banach lattice without a bibasic sequence and that this answers a question from Taylor–Troitsky. The decisive statement is Theorem 7.5:

$\phi(c_0) \subseteq FBL^{(p)}[c_0]$ does not contain any bibasic sequence whenever $1 \leq p < \infty$.

For $p = 1$, $FBL^{(1)}[c_0]$ is the ordinary free Banach lattice $FBL[c_0]$.

Why this answers the target. The map $\phi : c_0 \rightarrow FBL[c_0]$ is the canonical linear isometric embedding, so $\phi(c_0)$ is a closed infinite-dimensional subspace of the Banach lattice $FBL[c_0]$. Theorem 7.5 says this subspace contains no bibasic sequence. Hence the answer to Remark 4.4 of arXiv:1907.07589 is negative.

Corroborating pointer. E. Garcia-Sanchez, D. de Hevia, and P. Tradacete, *Free objects in Banach space theory*, arXiv:2302.10807, explicitly states that arXiv:1907.07589 Remark 4.4 asked whether there exists a subspace of a Banach lattice with no bibasic sequence, and that arXiv:2210.00614 Theorem 7.5 gives such a subspace by using the canonical copy of c_0 in $FBL[c_0]$.

Scope and limitations. This settles the general closed-subspace bibasic question from Remark 4.4. It does not address the separate uo-bibasic or order-continuous variants posed later in arXiv:1907.07589.

Search evidence. The run indexes were searched for arXiv:1907.07589, bibasic sequences, Remark 4.4, subspaces without bibasic sequences, and arXiv:2210.00614. No prior packet for this Taylor–Troitsky question was found. The local parsed sources for arXiv:2210.00614 and arXiv:2302.10807 contain the exact cross-references quoted above.